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Discussion Prompt
Quizzes
Graded Assignment: Regression to the Mean Submitted
Graded Assignment: Small Numbers Fallacy 10 min

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Next item

Regression to the Mean

Review Learning Objectives

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
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Assignment details

Due
Nov 6, 11:59 PM IST

Attempts
3 every 4 hours

Submitted
Nov 2, 12:46 AM IST

Your grade
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1. Suppose Mike is playing a game in which he throws three fair coins into the air (First Toss). For each "heads", he wins a dollar. Then, he tosses the three coins again (Second Toss). For each "heads" on this second toss, he also wins a dollar. (So he can win anywhere from $0 to $6 in one play of the game.)
What is the probability of tossing three heads on the first toss? (To three decimal places)
0.125
Correct — **1 / 1 point**

2. What is the probability of tossing 2 or more heads on the first toss? (To three decimal places)
0.875
Correct — **1 / 1 point**

3. What is the probability of tossing 0 heads on the first toss? (To three decimal places)
0.125
Correct — **1 / 1 point**

4. Suppose Mike tosses 0 heads on the first toss. What is the probability of doing better than that on the second toss? (To three decimal places)
0.875
Correct — **1 / 1 point**

5. Suppose Mike tosses 0 heads on the first toss and then curses the coins ("You stupid coins!") What is the probability of doing better on the second toss? (To three decimal places)
0.875
Correct — **1 / 1 point**

6. Suppose Mike tosses 3 heads on the first toss. What is the probability of doing worse than that on the second toss? (To three decimal places)
0.875
Correct — **1 / 1 point**

7. Suppose Mike tosses 3 heads on the first toss, and then praises the coins ("Good coins!") What is the probability of doing worse on the second toss? (To three decimal places)
0.875
Correct — **1 / 1 point**

8. Explain why these answers might lead Mike to (mistakenly) believe that criticizing the coins helps change their behavior in the right way, but praising the coins isn't a good idea.
Due to regression to mean effect
Feedback
This is an example of the "regression-to-the-mean" judgment phenomenon described in Tversky and Kahneman's paper.
1 / 1 point